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Experimental and kinetic modeling investigation of toluene combustion in premixed, one-dimensional and laminar toluene-oxygen-argon flames

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Abstract

One-dimensional, premixed and laminar toluene–oxygen–argon flames, with equivalence ratios of 0.7 and 1.0 were stabilized at low pressure on a flat flame burner. Mole fraction profiles of 18 chemical species have been measured by gas chromatography and molecular beam mass spectrometry. A kinetic model with a good predictive capability has been developed to compare with experimental data. A comparative reaction rate analysis of predicted flame structures has allowed us to determine main pathways of toluene combustion and their dependence on temperature and equivalence ratio.
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Keywords: Toluene; Flame; Oxidation; Kinetic; Model

1. Introduction

Nowadays, applied and fundamental investigations in the field of pollutant formation, alternative fuels or environmental impact of combustion products have become a central theme of research. One of the main goals in the area of hydrocarbon combustion is the understanding of polycyclic aromatic hydrocarbons (PAH) formation. This interest is due to their threat to human health [1]. Indeed, these heavy and highly stabilized aromatic molecules are carcinogenic and thought to be molecular intermediates in soot formation [2]. Toluene (C_7H_8), one of the smallest aromatic hydrocarbons, is considered as a PAH precursor in rich hydrocarbon flames. However, toluene is widely used as a fuel additive or in surrogate mixtures for gasoline due to its high anti-knocking capacity. Toluene can also be formed as a combustion intermediate and lead to PAH and soot formation when incomplete hydrocarbon oxidation occurs, as in diesel engines or several industrial processes. In order to reduce PAH and soot formation, a detailed understanding of chemical pathways involved in toluene oxidation under combustion conditions is necessary. Such a comprehension can be reached by the development of kinetic models able to reproduce experimental measurements.

Most of previous investigations into toluene combustion chemistry were mainly performed in chemical reactors and shock tubes. Venkat et al.

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[3], Emdee et al. [4], Brezinsky et al. [5] and Brezinsky [6] experimentally and numerically studied toluene oxidation at high temperature in a turbulent flow reactor and Dagaut et al. [7] and Bounaceur et al. [8] investigated C_7H_8 oxidation in a jet stirred reactor. Shock tubes experiments were also performed by several groups, including Bounaceur et al. [8], Sivaramakrishnan et al. [9,10], Pamidimukkala et al. [11] and Colket and Seery [12]. Surprisingly, toluene combustion has been poorly investigated in flames. Laminar flame speed measurements of toluene–air mixtures were performed by Davis et al. [13] and used to improve the kinetic model previously developed by Emdee et al. [4]. To our best knowledge, structure determinations of laminar, one-dimensional, and premixed pure toluene flames were only performed by Li et al. [14] and our group [15]. Both works describe an experimental analysis of such rich toluene-oxygen-argon flames and our previous study proposed a kinetic modeling analysis of our experimental measurements [15]. The mechanism included recent theoretical findings on toluene combustion chemistry, especially a pathway linking the benzyl radical (b-C_7H_7), a major intermediate in toluene combustion, to the cyclopentadienyl radical (C_5H_5) through fulvenallene (C_7H_6). The kinetic model had a good predictive capability and we demonstrated that the C_7H_6 pathway was one of the main toluene consumption route [15]. The present study provides experimental structures of two laminar, one-dimensional and premixed toluene-oxygen-argon flames, burning at low pressure with equivalence ratios of 1 and 0.7. The kinetic model previously developed for rich toluene combustion has been adapted for leaner conditions and its predictions are compared to experimental measurements.

2. Experimental methods

Mole fraction profiles of 18 chemical species have been measured by Gas Chromatography (GC) and Molecular Beam Mass Spectrometry (MBMS) in toluene-oxygen-argon flames stabilized on the burner of the experimental setup shown in Fig. 1. This setup has been described in a previous article [15]. The chromatographic separation is carried out by a MOLSIEVE 5A and a PORA PLOT Q columns (from Varian Inc.) for permanent gases of combustion and hydrocarbons, respectively. Quantitative detection is achieved by a Thermal Conductivity Detector (TCD) and a Flame Ionization Detector (FID) connected in series. Their responses are calibrated for permanent gases of combustion and a large range of stable hydrocarbons, including those observed into the flame. The calibration procedure and experimental errors have been discussed in a previous article [15].

Beside Gas Chromatography, Molecular Beam Mass Spectrometry (MBMS) has been used to measure mole fraction profiles of m/z ratios corresponding to C_7H_7 ($m/z = 90$) and C_7H_6O ($m/z = 106$). Indeed, these chemical species could not be identified by GC and were necessary to validate the kinetic model presented in the next section. The experimental setup used for MBMS measurements is similar to the one used for the Gas Chromatography analysis (Fig. 1) excepted that behind the quartz nozzle, samples are accelerated through differentially pumped chambers towards the ion source of a mass spectrometer. This setup and the calibration procedure have been extensively described in a previous article [15].

Temperature measurements at different heights of the flames have been accomplished by using a Pt/PtRh10% thermocouple, 0.1 mm in diameter, coated with a thin layer of Y_2O_3–BeO ceramic to prevent catalytic effects of Pt on chemical reactions occurring in the flame [16]. Data acquired have been corrected for radiation losses by the electrical compensation method.

3. Numerical simulation

The kinetic model developed in this study is an adaptation to stoichiometric and lean conditions of a mechanism previously validated against a rich toluene flame [15]. We ensured that the adapted model does not significantly change mole fraction profiles predicted in this previous study [15]. The

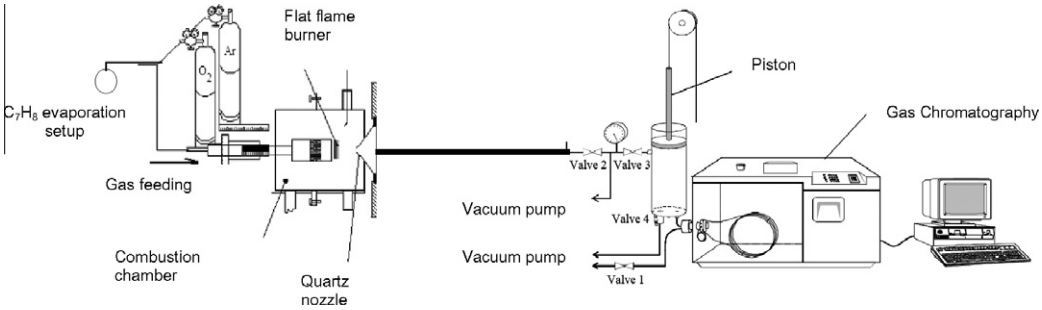

Fig. 1. Experimental setup used for hydrocarbons and permanent gases of combustion measurements.

model is mainly based on comprehensive kinetic mechanisms of Ristori et al. [17], Richter and Howard [18] and Konnov [19], and on theoretical studies of Cavalotti et al. [20]. The present mechanism is not definitive but it allows discussing some important findings which will be helpful for further development of a comprehensive and definitive model for aromatic oxidation. Main changes brought to the former mechanism are discussed further in the text. Calculations were performed by using the COSILAB software [21] and the experimental temperature profile has been used as an input parameter. The submodel used for C_5H_x to $C_{12}H_x$ chemistry along with its thermodynamic and transport data are provided as Supplementary data.

4. Experimental results and discussion

In the present study, two different one-dimensional, premixed and laminar toluene-oxygen-argon flames, with equivalence ratios of 0.7 (F07, see Table 1) and 1.0 (F10, see Table 1), were stabilized at 36 Torr on the flat flame burner of the experimental setup shown in Fig. 1. Besides F07 and F10, the experimental structure of a rich toluene-oxygen-argon flame (F20, see Table 1) described in a previous study [15] was used to validate the kinetic model over a large range of equivalence ratio.

Figure 2 shows experimental temperature profiles of F07, F10 and F20. F20 has the highest temperature profile and reaches 1743 K at 1.05 cm from the burner surface. F10 and F07 are more diluted flames (see Table 1) and have consequently a lower temperature than F20. Maximum temperatures of F10 and F07 respectively reach 1662 and 1630 K at 1.15 and 1.25 cm from the burner surface. Thus, in the present work the maximum flame temperature decreases with the equivalence ratio.

In the following discussion the predicted flame structure of F20 is not compared to experimental measurement as the developed kinetic model gives F20 predictions very close to those validated in an earlier work [15]. Nevertheless, the kinetic scheme of toluene combustion in F20 is discussed to allow a comparison with F10 and F07. Then, experimental flame structures of F10 and F07 are compared to data from the numerical simulation and kinetic schemes of these flames are described.

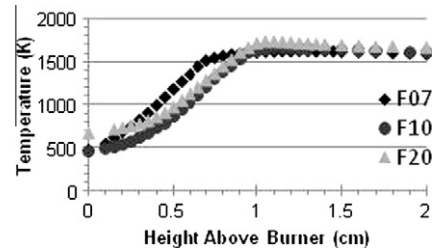

Fig. 2. Experimental flame temperatures.

Finally, the comparison of all kinetic schemes will allow us to determine the influence of temperature and equivalence ratio on toluene combustion chemistry.

4.1. Rich flame F20

Figure 3 shows that three main channels globally affect toluene combustion and Fig. 4 presents the kinetic scheme of F20. This kinetic scheme has been obtained by a comparative reaction rate analysis of the predicted flame structure. As shown in Fig. 4, 71% of C_7H_8 is converted by channel 1 (see Fig. 3) into benzyl radical (b-C_7H_7) through H abstraction reactions (see reactions (R1)–(R3)). Then, 63% of b-C_7H_7 decomposes into fulvenallene (C_7H_6) and H atom through reaction (R4). C_7H_6 reacts with H atom through reaction (R5) to produce the cyclopentadienyl radical (C_5H_5) and acetylene (C_2H_2). Beside this C_7H_6 route, 18% of b-C_7H_7 is oxidized to benzaldehyde (C_6H_5CHO) through reaction (R6). C_6H_5CHO is mainly consumed by H abstraction reactions on the aldehyde group yielding the C_6H_5CO radical. This radical dissociates into phenyl radical and CO through reaction (R7):

$$C_7H_8 + H \Longleftrightarrow \text{b-}C_7H_7 + H_2 \quad (R1)$$

$$C_7H_8 + OH \Longleftrightarrow \text{b-}C_7H_7 + H_2O \quad (R2)$$

$$C_7H_8 + O \Longleftrightarrow \text{b-}C_7H_7 + OH \quad (R3)$$

$$\text{b-}C_7H_7 \Longleftrightarrow C_7H_6 + H \quad (R4)$$

$$C_7H_6 + H \Longleftrightarrow C_5H_5 + C_2H_2 \quad (R5)$$

$$\text{b-}C_7H_7 + O \Longleftrightarrow C_6H_5CHO + H \quad (R6)$$

$$C_6H_5CO \Longleftrightarrow C_6H_5 + CO \quad (R7)$$

As shown in Fig. 4, channel 2 (see Fig. 3) dissociates 6% of toluene through reaction (R8) into phenyl and methyl radicals and a metathesis reaction

Table 1
Flame parameters.

Code	Equ. ratio	XC_7H_8	XO_2	XAr	Flow rate[a]
F07	0.7	2.3×10^{-2}	2.9×10^{-1}	6.9×10^{-1}	5.5
F10	1.0	2.3×10^{-2}	2.0×10^{-1}	7.7×10^{-1}	5.5
F20	2.0	9.9×10^{-2}	4.5×10^{-1}	4.6×10^{-1}	5.5

[a] Total gas flow rate in l/min at 298 K and 1 atm.

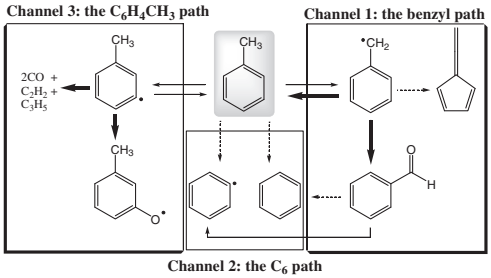

Fig. 3. Main channels of toluene consumption in investigated flames and their dependence on temperature and equivalence ratio (φ). Bold, dotted and regular arrows represent reactions which are respectively promoted, disadvantaged and not affected by a decrease of temperature or φ.

with H atom (see reaction (R9)) yields benzene and methyl radical. Then, benzene is converted into phenyl radical (C_6H_5) through H abstraction reactions with H, OH and O. Kumaran and Michael found that C_6H_5 radical reacts with O_2 through two major channels yielding either O atom or H atom [22]. They supposed that byproducts of these O and H atoms pathways were phenoxy radical (C_6H_5O) and benzoquinone ($C_6H_4O_2$), respectively (see reactions (R10) and (R11)):

$$C_7H_8 \Longleftrightarrow C_6H_5 + CH_3 \quad (R8)$$

$$C_7H_8 + H \Longleftrightarrow C_6H_6 + CH_3 \quad (R9)$$

$$C_6H_5 + O_2 \Longleftrightarrow C_6H_5O + O \quad (R10)$$

$$C_6H_5 + O_2 \Longleftrightarrow C_6H_4O_2 + H \quad (R11)$$

$$C_6H_5 + O_2 \Longleftrightarrow 2CO + C_2H_2 + C_2H_3 \quad (R12)$$

$$C_6H_5O \Longleftrightarrow C_5H_5 + CO \quad (R13)$$

To our best knowledge benzoquinone was never observed in significant concentration in benzene flames or other experimental studies devoted to benzene oxidation. Consequently, in our kinetic model reaction (R11) has been substituted with the overall reaction (R12) that opens the phenyl ring giving 2CO, C_2H_2 and C_2H_3. Since the vinyl radical (C_2H_3) mainly dissociates into C_2H_2 and H atom, reaction (R12) produces H atom as reaction (R11). C_6H_5O dissociates through reaction (R13) into CO and C_5H_5.

Figure 4 shows that 12% of toluene is converted by channel 3 (see Fig. 3) into ($C_6H_4CH_3$)

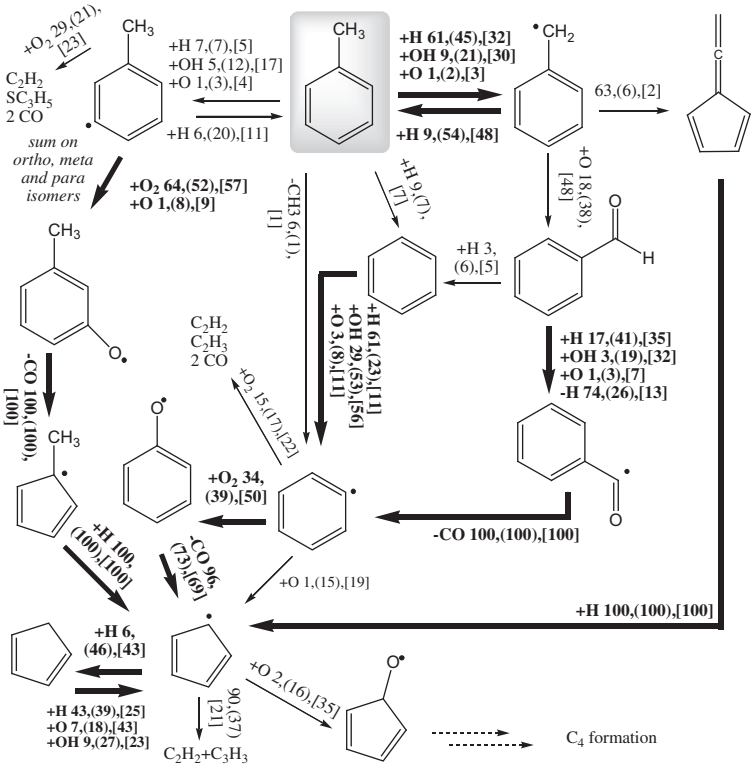

Fig. 4. Reaction contributions to consumption of chemical species involved in main pathways of F20, F10 and F07. Bold arrows indicate main consumption pathways in F10 and F07. Numbers above arrows indicate percentages of reaction contributions to the consumption of each compound in F20, (F10) and [F07], respectively.

by H abstraction reactions with H, OH and O (see reactions (R14)–(R16)). Whereas our previous study [15] only considered H abstraction reactions on the methyl group of toluene, those on the phenyl ring giving $C_6H_4CH_3$ have been included in the present work. Kinetic parameters of reactions (R14)–(R16) where respectively taken from works of Ellis et al. [23], Seta et al. [24] and estimated by analogy with C_6H_6:

$$C_7H_8 + H \Longleftrightarrow C_6H_4CH_3 + H_2 \quad (R14)$$

$$C_7H_8 + OH \Longleftrightarrow C_6H_4CH_3 + H_2O \quad (R15)$$

$$C_7H_8 + O \Longleftrightarrow C_6H_4CH_3 + OH \quad (R16)$$

Reaction pathways involving the $C_6H_4CH_3$ radical and their kinetic parameters have been added in the mechanism by analogy with reactions of the C_6H_5 radical. The $C_6H_4CH_3$ radical reacts with O_2 yielding mainly the $OC_6H_4CH_3$ radical (reaction (R17)). A minor route produces 2CO, C_2H_2 and 1-methylvinyl radical (C_3H_5, $CH_2C(\cdot)CH_3$) through reaction (R18). As vinyl radical, C_3H_5 is an H atom precursor since it dissociates into propyne (PC_3H_4) and H atom.

$$C_6H_4CH_3 + O_2 \Longleftrightarrow OC_6H_4CH_3 + O \quad (R17)$$

$$C_6H_4CH_3 + O_2 \Longleftrightarrow 2CO + C_2H_2 + C_3H_5 \quad (R18)$$

$$OC_6H_4CH_3 \Longleftrightarrow C_5H_4CH_3 + CO \quad (R19)$$

$$C_5H_4CH_3 + H \Longleftrightarrow C_5H_5 + CH_3 \quad (R20)$$

$OC_6H_4CH_3$ dissociates through reaction (R19) into the methylcyclopentadienyl radical ($C_5H_4CH_3$) and CO, in the same way as C_6H_5O gives C_5H_5 and CO. Then, $C_5H_4CH_3$ is consumed through the metathesis reaction (R20) giving the cyclopentadienyl radical (C_5H_5) and CH_3. Since ortho, meta and para isomers of $OC_6H_4CH_3$ radical should all give resonant forms of $C_5H_4CH_3$ radical, the kinetic model does not differentiate between isomers of methylphenoxy and methylphenyl radicals. In the present mechanism, $C_6H_4CH_3$ and $OC_6H_4CH_3$ represent the sum on ortho, meta and para positions.

As shown in Fig. 4, the three main channels of toluene consumption converge towards the C_5H_5 radical formation. 90% of C_5H_5 dissociates into acetylene (C_2H_2) and propargyl radicals (C_3H_3), 6% recombines with H atom yielding cyclopentadiene (C_5H_6) and 2% is oxidized to the cyclopentadienoxy radical (C_5H_5O). C_5H_5O dissociates into CO and C_4H_5, an important precursor of C_4H_x hydrocarbons. CH_4 is mainly produced from CH_3, which is a major product of channels 2 and 3.

4.2. Stoichiometric and lean flames F10 and F07

Figures 5 and 6 present experimental and predicted mole fraction profiles of F10 and F07, respectively. They show that the developed kinetic model generally well reproduces the experimental flame structures of both F10 and F07. Toluene is completely consumed in both flames, whereas oxygen mole fractions of 2.5×10^{-2} and 9.0×10^{-2} remain in the burnt gases of F10 and F07, respectively. The lowest equivalence ratio of F07 explains the highest oxygen persistence in its burnt gases. In F10 and F07, maximum mole fractions of CO_2 are higher than those of CO, whereas the opposite is observed in F20. This can be explained by the highest OH concentration in stoichiometric and lean flames, which promotes the main CO_2 formation path: $CO + OH \Leftrightarrow CO_2 + H$. Figure 4 presents reaction contributions to the consumption of chemical species in F10 and F07. In these flames, toluene consumption occurs through the same three pathways than in F20 (see Fig. 3). Whereas these channels are similar, the relative contributions of their reactions are different in F10 and F07, compared to F20. Indeed, F10 and F07 have both lower temperatures and H atom concentrations than F20 due to their highest dilution ratio and their smallest equivalence ratio, respectively (see Table 1). Consequently, highly temperature dependent reactions and those involving H atom as a reactant are disadvantaged in F10 and F07, compared to F20. On the contrary, reactions with O atom and OH radical are promoted in F10 and F07 due to the highest concentration of oxygenated compounds, compared to F20. The following discussion compares the kinetic scheme of F20 (Fig. 4) to those of F10 and F07 (Fig. 4) in order to determine how temperature and equivalence ratio influence the chemistry of toluene combustion. Results are summarized in Fig. 3 where dotted, bold and regular arrows represent chemical pathways which are respectively disadvantaged, promoted and not affected by a decrease of temperature and equivalence ratio.

Entrance reactions of channel 1 are H abstraction reactions with H atom, O atom and OH radical (see Fig. 4 and reactions (R1)–(R3)). Since in F10 and F07, compared to F20, reactions with H atom are disadvantaged, whereas those with O atom and OH radical are promoted, the overall contribution of channel 1 to toluene consumption should not be highly affected by the drop of the equivalence ratio. Indeed, in F10 and F07, the percentage of toluene which enters into channel 1 (about 65%, see Fig. 4) is close to the one in F20 (71%, see Fig. 4). 54% of benzyl radical produced by channel 1 in F10 is converted back into toluene through the recombination reaction (R21), whereas it was essentially isomerized into C_7H_6 in F20 through the reaction (R4) (see Fig. 4). Therefore, in F10 and F07 the net contribution of channel 1 to toluene consumption drops, compared to F20:

$$b\text{-}C_7H_7 + H \Longleftrightarrow C_7H_8 \quad (R21)$$

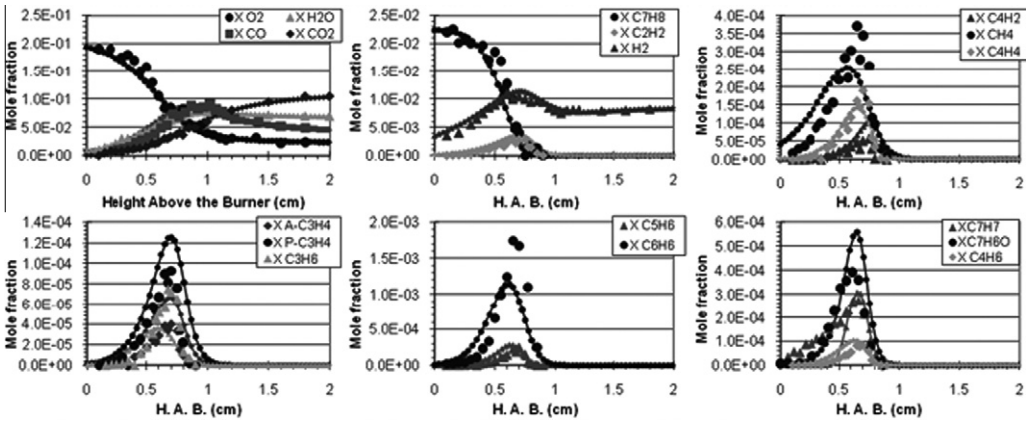

Fig. 5. Experimental (symbols) and numerical (soild curves) flame structures of F10.

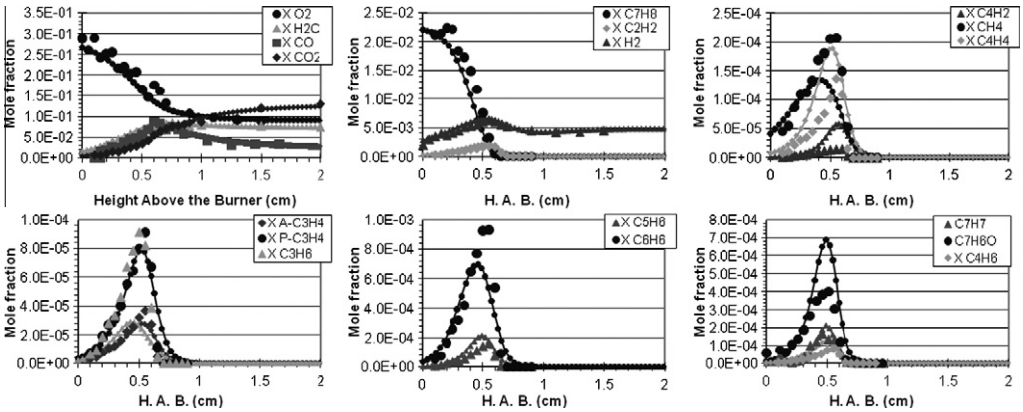

Fig. 6. Experimental (symbols) and numerical (soild curves) flame structures of F07.

This observation can be explained by the large difference between high and close to zero activation energies of isomerization reaction (R4) and free radical recombination reaction (R21), respectively. Indeed, it deeply modifies their branching ratio at lower flame temperatures than those measured in F20. Although the flame temperature still decreases in F07, only 48% of benzyl radical recombines with H atom into toluene, compared to 54% in F10. This can be explained either by the promotion of the benzyl oxidation reaction (R6) or by the drop in H atom concentration, decreasing the speed of reaction (R21). Compared to F20, the highest O atom concentration of F10 and F07 increases the contribution of the oxidation reaction (R6). Indeed, it respectively consumes 38% and 48% of benzyl radical in F10 and F07, compared to 18% in F20. Therefore, as temperature and equivalence ratio decreases, a larger ratio of benzyl radical is converted back to toluene or leaves channel 1 for channel 2 through reactions (R6) and (R7) (see Figs. 3 and 4). In F10 and F07, the present kinetic model well reproduces mole fraction profiles performed by MBMS at m/z ratios corresponding to benzyl radical and benzaldehyde (see Figs. 5 and 6). Measurement of the mole fraction profile corresponding to the m/z ratio of C_7H_6 was beyond the detection limit of our mass spectrometer. This observation agrees with the very low C_7H_6 mole fractions predicted by the kinetic model in F10 and F07: 8.1×10^{-6} and 3.5×10^{-6}, respectively.

As shown in Fig. 4, channel 2 involves both a metathesis reaction with an H atom (reaction (R8)) and a dissociation reaction with an high activation energy causing a highly temperature dependent behavior (reaction (R9), $k_9 = 5.0 \times 10^{16} \exp(-49{,}321/T)\ s^{-1}$ [25]). Therefore, this channel is greatly disadvantaged as the equivalence ratio and the flame temperature decrease. As a result, its contribution to toluene consumption respectively falls to 1% in F10 and F07, compared to 6% in F20. Experimental mole

fractions of benzene are well predicted by the kinetic model (see Figs. 5 and 6).

Entrance reactions of channel 3 (see (R14)–(R16)) are H abstraction reactions with H atom, O atom and OH radical. Then, in the same way as channel 1 the overall contribution of channel 3 to toluene consumption is not highly affected by the drop of the equivalence ratio. However, in F10 and F07, the percentage of toluene entering into channel 3 (22% and 26%, respectively; see Fig. 4) is more important than in F20 (13%, see Fig. 4). This phenomenon is due to the decrease of the contribution of channel 2 to the toluene consumption. Methyl radical (CH_3) is a major product of channels 2 and 3 and the main precursor of CH_4, well predicted in F10 and F07 (see Figs. 5 and 6). 1-Methylvinyl radical (C_3H_5, $CH_2C(\cdot)CH_3$) is mainly formed through channel 3 and is responsible of propyne (P-C_3H_4) production, well predicted in F10 and F07 (see Figs. 5 and 6). In F10, the kinetic model underestimates C_3H_6 and allene (A-C_3H_4), also underestimated in F07 (see Figs. 5 and 6). Since they are well predicted in F20, it should be explained by uncertainties in oxidation reactions of the C_3H_x submechanism. Figure 4 shows that in F10 and F07 channels 1, 2 and 3 converge towards the cyclopentadienyl radical formation, as in F20. In F10 and F07, the cyclopentadienyl radical is consumed through the same 3 main pathways than in F20: a dissociation reaction into C_2H_2 and C_3H_3 and recombination reactions with H and O into C_5H_6 and C_5H_5O, respectively (see Fig. 4). In F10 and F07 C_5H_6 is accurately predicted by the kinetic model and C_5H_5O is a precursor of C_4H_x hydrocarbons which are generally well predicted, too (see Figs. 5 and 6).

5. Conclusions

The present work fills the lack of experimental data on one-dimensional, premixed and laminar pure toluene flames by providing experimental mole fractions profiles of 18 chemical species measured by Gas Chromatography and Molecular Beam Mass Spectrometry in a stoichiometric and a lean toluene-oxygen-argon flames.

The kinetic model of a former study devoted to rich toluene combustion has been adapted to account for stoichiometric and lean conditions. It reproduces with a good predictive capability mole fraction profiles of most of the measured chemical species over a range of equivalence ratio spreading from 0.7 to 2. Main toluene combustion pathways have been determined by an extensive flow rate analysis of predicted data. Over the wide range of equivalence ratio the toluene consumption is performed through three major channels: the benzyl radical production pathway, the C_6 formation route and the methylphenyl radical channel. The three pathways converge towards the cyclopentadienyl radical formation. In high temperature and rich toluene combustion, the C_6 formation route and the benzyl radical pathway are promoted, whereas these channels are disadvantaged in flames with lower temperature and equivalence ratio. As a result, the methylphenyl route becomes more important when analyzing stoichiometric or lean toluene flames and should be taken into account in kinetic mechanisms. The developed kinetic model will help further studies on toluene oxidation as well as on PAH and soot formation.

Acknowledgment

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.151.

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